SANATH KOUNDINYA

Roll No. 250973 | B.Tech, Electrical Engineering
Indian Institute of Technology Kanpur

sanathk25@iitk.ac.in
+91 97405 00712
GitHub: github.com/Sanath-Koundinya

ACADEMIC QUALIFICATIONS

Year	Degree / Examination	Institute	CPI / %
2025 – 2029	B.Tech, Electrical Engineering (3 A* grades)	Indian Institute of Technology Kanpur	CPI 9.63 / 10
2025	CBSE Class XII (PCM + CS)	Alpine Public School, Bengaluru	JEE Adv. AIR 1063
2023	CBSE Class X	BGS National Public School, Bengaluru	99.5% (State Topper)

RESEARCH EXPERIENCE

Summer Research Intern, WSAIS, IIT Kanpur | Robot learning (MimicGen / SkillGen) *Summer 2026*

- Implemented the MimicGen data-generation pipeline (basis of SkillMimicGen, CoRL 2024) on robosuite and robomimic, and modified the codebase to scale automated demonstration generation from a handful of human demos.
- Generated 200 rollouts on the peg-in-hole assembly task (92 verified successes from 10 source demos); trained behaviour-cloning policies and built before/after rollout comparisons with playback videos.
- Moved on to the SkillGen implementation in Isaac Sim / Isaac Lab on the lab workstation; prepared a research presentation on the SkillMimicGen paper.

Undergraduate Project (UGP): Computational Photography | Canonical face and albedo from video *Aug 2026 – Present*

- Built a colour-correction validation: SpyderCheckr-48 chart, sRGB→XYZ→Lab ΔE. Mean ΔE fell from 15.63 to 11.69 (25.2% lower error), with 37 of 48 patches improved. Fixed and extended my own automated LUT pipeline (ffmpeg, OpenCV).
- Built a FLAME 3D face-fitting pipeline (analysis-by-synthesis, Adam, two-stage shape/expression fit); fixed "shape explosion" by MAP-style regularisation, cutting parameter magnitudes from 9.6 / 48.6 to ≤ 3.
- Ran MICA, metrical-tracker, FlashAvatar and RGBAvatar end to end; patched a CUDA out-of-memory failure (8.28 GiB detector allocation) that blocked FLAME extraction for a 31,686-video dataset, validated by loss, bounding-box and parameter comparisons.

PROJECTS

Hardware L2 Order Book in SystemVerilog (FPGA / HDL, low-latency trading) github.com/Sanath-Koundinya/hw-orderbook

- 64 price levels per side, 1-cycle message updates, best bid/ask from occupancy masks; verified in cocotb with 5000 random messages against a Python reference model; Verilator lint clean.
- Yosys synthesis: flip-flops cut from 2176 to 128 by resetting only the masks, letting quantities map to RAM. Simulation and synthesis only so far; ITCH-style parser planned.

Project FOCI: FPGA Camera, Electronics Club | Summer project, Verilog/VHDL on EDGE Spartan-7 *May – Aug 2026*

- Mentee on a from-scratch FPGA camera: clock dividers and strobe enables, UART transmitter, I2C-style sensor master with testbenches; board exercises (VHDL LCD, ST7735 TFT driver, 3×3 correlation) in Vivado, plus a live OV7670-to-VGA camera demo on hardware.

EyeSeeYou: Smart Surveillance (Raspberry Pi, YOLO, OpenCV, OCR)

- Real-time licence-plate recognition with a multithreaded capture, inference, AES-256 encryption and MongoDB GridFS pipeline.

SCHOLASTIC & MATHEMATICAL ACHIEVEMENTS

- **Estimathon:** 4th place (team of two); top Y25 team in our round. **RMO:** qualified twice; IOQM Certificate of Merit; NMTC Stage 2 (2024); Aryabhatta Ganit Challenge.
- **JEE Advanced AIR 1063** (top 0.1%), JEE Main AIR 3879; AP Calculus BC and CS A 5/5; SAT 1530. Math Olympiads: UIMO National Rank 30, SOF IMO International Rank 33 / 60; NSEP Merit; SOF NSO Rank 15.
- **1st (joint 3rd), Electrovista Hackathon** and **Winner, AstroVenture** (pan-IITK, Electronics and Astronomy Clubs).

TECHNICAL SKILLS & LEARNING

- **HDL / Hardware:** SystemVerilog, Verilog, VHDL, cocotb, Verilator, Yosys, Vivado. **Programming:** Python, C, PyTorch, OpenCV, Git, Linux. **Math:** Linear Algebra, Probability, Discrete Maths, Differential Equations.
- **Learning Undertaking:** Completing free certifications in digital design, FPGA prototyping and low-latency systems, and building more SystemVerilog/VHDL projects on GitHub. Eager to learn any tool or domain a role requires.

POSITIONS OF RESPONSIBILITY

- **Secretary, Debate Society** (Jan 2025 – Present; represented IITK at NSMPD, Dialectica, VIT Pre-UADC, Charcha); **Secretary, Book Club** (Aug 2025 – Present); **Core Group, Vox Populi** (2026); **Founding Member, Animal Allies** (2025).

EXTRA-CURRICULAR

- **Diploma in Carnatic Music, Chandigarh University**; Grade A Double Star, KalaSourabha 2023; Finalist, Jhankriti 2023; High Distinction 96.5% (Karnataka Board). Volunteer, Ramakrishna Mission (since 2022). Languages: English (IELTS 8.5), Hindi, Kannada, Sanskrit.